

DICKINSON ROOSEVELT, ND, USA

WMO: 727630

Lat: 46.799N

Lon: 102.797W

Elev: 2580

StdP: 13.38

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24012

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-11.1	-23.6	1.6	-16.5	-17.5	2.3	-10.7	33.4	17.8	30.5	19.7	11.6	290	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.6	93.0	66.3	89.5	66.2	85.7	65.2	71.7	84.0	69.5	82.4	67.7	80.3	12.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.9	113.1	78.1	65.5	103.8	75.5	63.4	96.1	73.2	37.2	83.9	35.1	82.6	33.5	79.8	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.0	26.6	24.1		-23.0	99.8	4.7	3.5	-26.4	102.3	-29.1	104.3	-31.7	106.3	-35.1	108.8
				-23.4	75.5	4.6	2.4	-26.7	77.2	-29.4	78.5	-31.9	79.9	-35.2	81.6
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	16.3	18.9	29.6	41.5	52.5	62.3	69.9	68.5	58.8	44.2	30.5	19.4	
	DBStd	21.83	14.79	14.40	13.93	10.36	8.45	6.91	6.15	6.69	9.05	10.05	12.26	13.84	
	HDD50	4705	1045	870	637	285	72	4	0	0	24	233	587	948	
	HDD65	8507	1510	1290	1096	705	394	127	23	42	226	647	1034	1413	
	CDD50	2096	0	0	6	31	150	374	616	574	288	53	4	0	
	CDD65	422	0	0	0	0	7	47	175	151	40	2	0	0	
	CDH74	5849	0	0	3	32	170	647	2198	2032	706	60	1	0	
	CDH80	2373	0	0	0	6	36	184	950	908	275	14	0	0	
Wind	WSAvg	12.0	13.0	12.4	12.7	12.8	12.7	11.4	10.2	10.6	11.3	12.2	12.2	12.5	
Precipitation	PrecAvg	16.1	0.4	0.4	0.7	1.9	2.2	3.4	2.3	1.4	1.6	0.9	0.5	0.4	
	PrecMax	25.4	1.0	2.3	2.5	4.8	5.7	7.2	5.7	4.0	6.3	5.4	2.7	1.1	
	PrecMin	10.6	0.0	0.0	0.1	0.1	0.1	0.6	0.4	0.0	0.2	0.0	0.0	0.0	
	PrecStd	3.6	0.3	0.4	0.7	1.3	1.3	1.6	1.4	1.2	1.3	1.1	0.6	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	55.1	70.4	78.5	84.3	90.3	99.6	97.8	94.0	81.0	66.7	52.6	
		MCWB	40.5	42.6	50.2	55.0	61.0	68.4	67.9	67.2	64.4	57.1	49.1	41.5	
	2%	DB	42.9	47.1	61.3	70.5	78.8	85.0	93.1	92.9	87.4	72.8	59.0	45.8	
		MCWB	36.3	38.5	45.7	51.8	59.7	66.2	67.2	65.8	62.8	54.7	45.0	37.6	
	5%	DB	38.9	41.4	54.9	64.5	74.1	81.2	89.4	89.5	82.1	66.0	52.4	40.7	
		MCWB	34.2	35.7	43.4	48.8	56.6	65.0	67.5	65.5	60.6	50.7	41.6	34.4	
	10%	DB	36.0	37.1	48.8	59.9	69.8	77.3	85.2	85.2	76.9	60.4	46.7	36.8	
		MCWB	32.3	33.0	40.2	46.3	55.4	62.5	66.9	64.5	58.9	48.3	38.7	32.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	44.2	51.5	56.7	65.1	73.9	75.7	73.1	68.5	59.5	49.9	42.2
			MCDB	50.0	53.7	66.5	74.4	76.8	84.9	86.3	86.7	84.1	77.1	65.8	51.2
2%		WB	37.0	39.3	47.5	53.0	61.5	69.5	72.5	70.0	65.7	55.7	45.9	38.0	
		MCDB	42.2	46.4	60.5	68.7	74.6	79.6	84.1	83.8	79.8	68.0	57.7	44.8	
5%		WB	34.6	36.0	44.0	50.1	58.8	66.9	70.4	68.1	63.2	52.5	42.2	35.0	
		MCDB	38.7	41.1	54.4	62.6	70.6	77.5	83.2	81.9	76.7	64.3	51.1	40.4	
10%		WB	32.3	33.4	40.5	47.4	56.3	64.4	68.5	66.3	60.3	49.5	39.3	32.5	
		MCDB	35.8	36.8	48.5	58.4	67.2	74.6	81.2	80.1	74.2	59.3	46.3	36.7	
Mean Daily Temperature Range		5% DB	MDBR	18.2	18.7	20.9	23.6	23.5	23.5	26.6	28.0	26.3	22.5	20.2	17.8
			MCDBR	20.0	23.2	30.9	32.9	31.2	29.1	32.2	33.7	34.8	32.5	29.0	22.2
	MCWBR		15.1	15.9	18.0	17.8	15.6	14.2	12.8	12.9	14.8	17.0	17.3	15.4	
	5% WB	MCDBR	19.4	22.0	29.9	30.7	26.8	26.3	27.2	28.9	30.2	29.5	27.7	21.9	
		MCWBR	15.0	15.5	18.0	17.2	14.4	14.6	13.5	12.9	13.8	15.9	17.0	15.8	
Clear-Sky Solar Irradiance	taub	0.252	0.258	0.276	0.317	0.339	0.350	0.359	0.368	0.325	0.295	0.263	0.248		
	taud	2.500	2.524	2.535	2.424	2.421	2.426	2.433	2.376	2.476	2.544	2.526	2.511		
	Ebn at Noon	276	298	305	298	293	290	285	278	283	278	269	265		
	Edh at Noon	20	24	28	34	35	36	35	35	29	23	19	17		
All-Sky Solar Radiation	RadAvg	523	836	1197	1557	1842	2083	2198	1813	1374	874	552	399		
	RadStd	41	84	92	91	166	157	88	121	102	95	65	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page